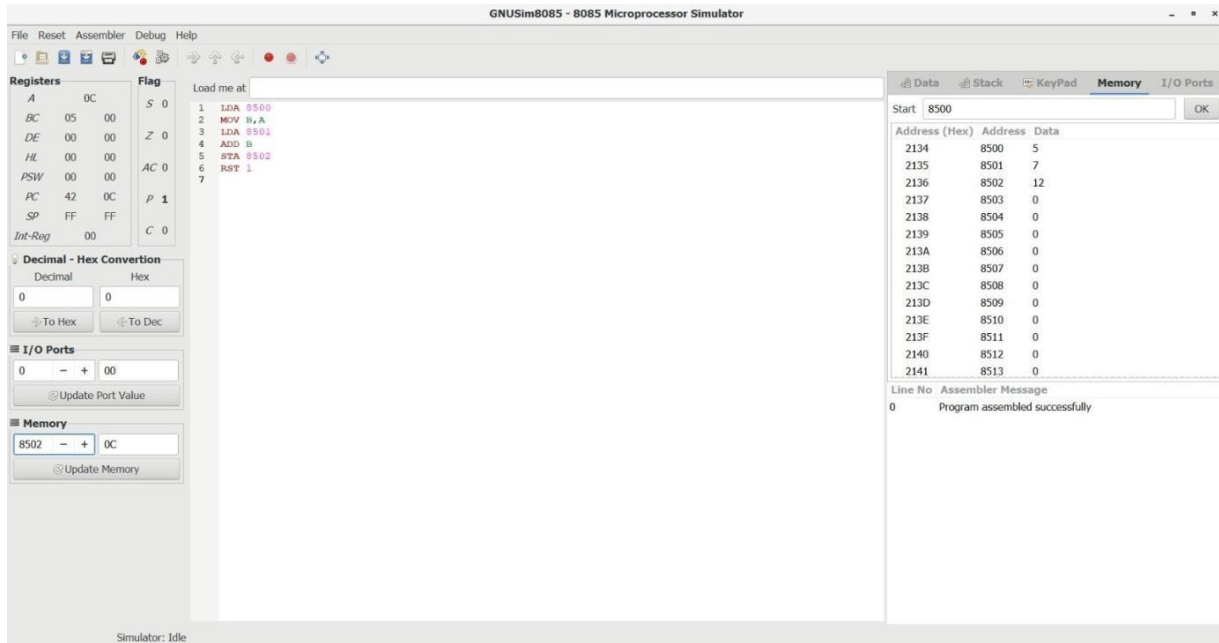
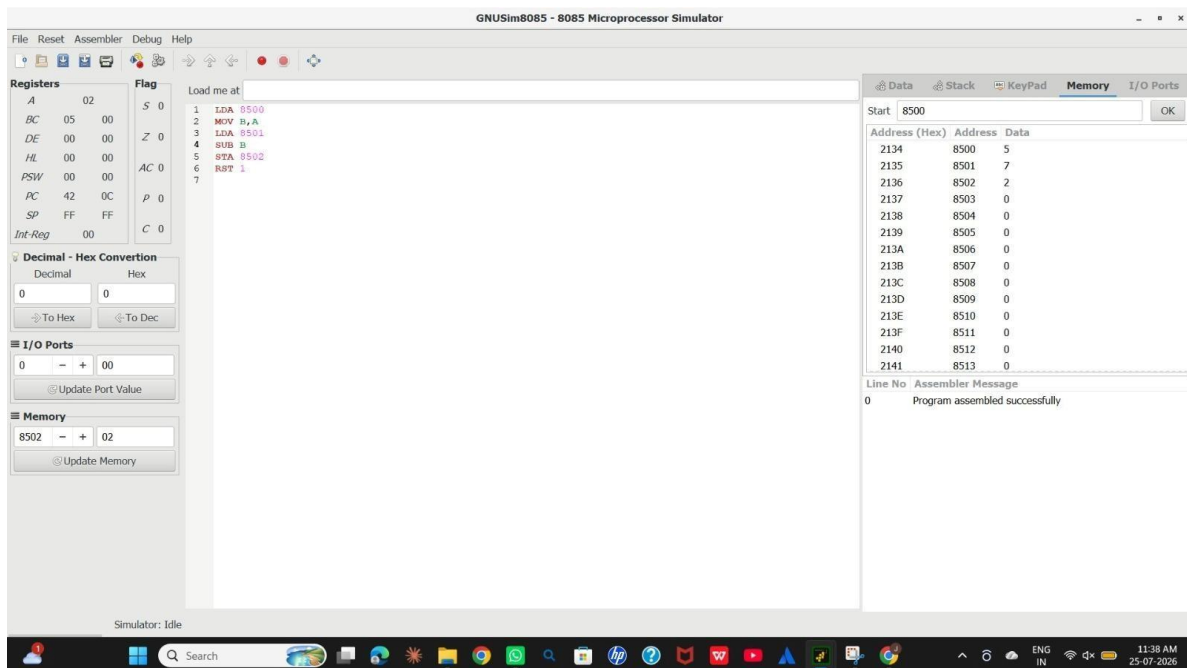


1) 8 Bit Addition



2) 8 Bit Subtraction



3) 8 Bit Add with carry

The screenshot shows the GNUSim8085 - 8085 Microprocessor Simulator interface. The assembly code in the center is as follows:

```
1 LDA 2500
2 MOV B,A
3 LDA 2501
4 MOV C,A
5 MVI A,00
6 LOOP: ADD B
7 DCR C
8 JNZ LOOP
9 STA 2502
10 HLT
```

The Registers window on the left shows the following values:

Register	Value
A	32
BC	0A 00
DE	00 00
HL	00 00
PSW	00 00
PC	42 13
SP	FF FF
Int-Reg	00

The Flag window shows: S 0, Z 1, AC 0, P 1, C 0.

The Memory window on the right shows the following data:

Address (Hex)	Address	Data
09C4	2500	10
09C5	2501	5
09C6	2502	50
09C7	2503	0
09C8	2504	0
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

The I/O Ports window shows: 0 - + 00.

The Memory window shows: 2501 - + 05.

The Assembler Message window shows: 0 Program assembled successfully.

4) 8 Bit Sub with Borrow

The screenshot shows the GNUSim8085 - 8085 Microprocessor Simulator interface. The assembly code in the center is as follows:

```
1 LDA 2500
2 MOV B,A
3 LDA 2501
4 MOV C,A
5 MVI D,00
6 MOV A,B
7 LOOP: SUB C
8 JC STOP
9 INR D
10 JMP LOOP
11 STOP: ADD C
12 STA 2503
13 MOV A,D
14 STA 2502
15 HLT
```

The Registers window on the left shows the following values:

Register	Value
A	05
BC	0A 02
DE	05 00
HL	00 00
PSW	00 00
PC	42 1C
SP	FF FF
Int-Reg	00

The Flag window shows: S 0, Z 1, AC 1, P 1, C 1.

The Memory window on the right shows the following data:

Address (Hex)	Address	Data
09C4	2500	10
09C5	2501	2
09C6	2502	5
09C7	2503	0
09C8	2504	0
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

The I/O Ports window shows: 0 - + 00.

The Memory window shows: 2501 - + 02.

The Assembler Message window shows: 0 Program assembled successfully.

5) 8 Bit Multiplication

The screenshot shows the GNUSim8085 - 8085 Microprocessor Simulator interface. The assembly code is as follows:

```
1 LDA 8500
2 MOV B,A
3 LDA 8501
4 SUB B
5 STA 8502
6 RST 1
7
```

The registers and flags are displayed on the left. The memory window on the right shows the following data:

Address (Hex)	Address	Data
2134	8500	5
2135	8501	7
2136	8502	2
2137	8503	0
2138	8504	0
2139	8505	0
213A	8506	0
213B	8507	0
213C	8508	0
213D	8509	0
213E	8510	0
213F	8511	0
2140	8512	0
2141	8513	0

The simulator status bar at the bottom indicates "Simulator: Idle".

6) Division with Quotient

The screenshot shows the GNUSim8085 - 8085 Microprocessor Simulator interface. The assembly code is as follows:

```
1 LDA 2500
2 MOV B,A
3 LDA 2501
4 CMP B
5 JC FIRST
6 STA 2502
7 HLT
8 FIRST: MOV A,B
9 STA 2502
10 HLT
```

The registers and flags are displayed on the left. The memory window on the right shows the following data:

Address (Hex)	Address	Data
09C4	2500	10
09C5	2501	20
09C6	2502	20
09C7	2503	0
09C8	2504	0
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

The simulator status bar at the bottom indicates "Simulator: Idle".

7) Division without Quotient

The screenshot shows the GNUSim8085 - 8085 Microprocessor Simulator interface. The Registers window displays the following values:

Register	Value
A	14
BC	0A 00
DE	00 00
HL	00 00
PSW	00 00
PC	42 0F
SP	FF FF
Int-Reg	00

The Flag window displays the following values:

Flag	Value
S	0
Z	0
AC	0
P	1
C	0

The Memory window shows the following data:

Address (Hex)	Address	Data
09C4	2500	10
09C5	2501	20
09C6	2502	20
09C7	2503	0
09C8	2504	0
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

The Assembler Message window shows the following message:

```
Line No  Assembler Message
0         Program assembled successfully
```

8) Greatest Of 2 Numbers

The screenshot shows the GNUSim8085 - 8085 Microprocessor Simulator interface after the execution of the 'Greatest Of 2 Numbers' program. The Registers window displays the following values:

Register	Value
A	14
BC	14 00
DE	00 00
HL	00 00
PSW	00 00
PC	42 14
SP	FF FF
Int-Reg	00

The Flag window displays the following values:

Flag	Value
S	1
Z	0
AC	0
P	0
C	1

The Memory window shows the following data:

Address (Hex)	Address	Data
09C4	2500	20
09C5	2501	15
09C6	2502	20
09C7	2503	0
09C8	2504	0
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

The Assembler Message window shows the following message:

```
Line No  Assembler Message
0         Program assembled successfully
```

9) Smallest Of 2 Numbers

The screenshot shows the GNUSim8085 - 8085 Microprocessor Simulator interface. The assembly code is as follows:

```
1 LDA 2500
2 MOV B,A
3 LDA 2501
4 CMP B
5 JC SECOND_SMALLER
6 MOV A,B
7 STA 2502
8 HLT
9 SECOND_SMALLER: STA 2502
10 HLT
```

The registers window shows the following values:

Register	Value
A	05
BC	1E 00
DE	00 00
HL	00 00
PSW	00 00
PC	42 14
SP	FF FF
Int-Reg	00

The flag window shows the following values:

Flag	Value
S	1
Z	0
AC	0
P	1
C	1

The memory window shows the following values:

Address (Hex)	Address	Data
09C4	2500	30
09C5	2501	5
09C6	2502	5
09C7	2503	0
09C8	2504	0
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

The assembler message window shows the following message:

```
Line No Assembler Message
0 Program assembled successfully
```

10) Swapping

The screenshot shows the GNUSim8085 - 8085 Microprocessor Simulator interface. The assembly code is as follows:

```
1 LDA 2500
2 MOV B,A
3 LDA 2501
4 MOV C,A
5 MOV A,B
6 STA 2501
7 MOV A,C
8 STA 2500
9 HLT
```

The registers window shows the following values:

Register	Value
A	28
BC	32 28
DE	00 00
HL	00 00
PSW	00 00
PC	42 11
SP	FF FF
Int-Reg	00

The flag window shows the following values:

Flag	Value
S	1
Z	0
AC	0
P	1
C	1

The memory window shows the following values:

Address (Hex)	Address	Data
09C4	2500	40
09C5	2501	50
09C6	2502	5
09C7	2503	0
09C8	2504	0
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

The assembler message window shows the following message:

```
Line No Assembler Message
0 Program assembled successfully
```

11) 1s Complement

The screenshot shows the GNUSim8085 - 8085 Microprocessor Simulator interface. The assembly code in the center is as follows:

```
1 LDA 2500
2 CMA
3 STA 2501
4 HLT
```

The registers on the left show the following values:

Register	Value
A	E3
BC	32 28
DE	00 00
HL	00 00
PSW	00 00
PC	42 08
SP	FF FF
Int-Reg	00

The flags on the right show: S 1, Z 0, AC 0, P 1, C 1.

The memory window on the right shows the following data:

Address (Hex)	Address	Data
09C4	2500	28
09C5	2501	227
09C6	2502	5
09C7	2503	0
09C8	2504	0
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

The assembler message at the bottom right states: "Program assembled successfully".

12) 2s Complement

The screenshot shows the GNUSim8085 - 8085 Microprocessor Simulator interface. The assembly code in the center is as follows:

```
1 LDA 2500
2 ANI 01H
3 JZ EVEN
4 MVI A, 01H
5 STA 2501
6 HLT
7 EVEN: MVI A, 00H
8 STA 2501
9 HLT
```

The registers on the left show the following values:

Register	Value
A	00
BC	32 28
DE	00 00
HL	00 00
PSW	00 00
PC	42 14
SP	FF FF
Int-Reg	00

The flags on the right show: S 0, Z 1, AC 1, P 1, C 0.

The memory window on the right shows the following data:

Address (Hex)	Address	Data
09C4	2500	24
09C5	2501	0
09C6	2502	5
09C7	2503	0
09C8	2504	0
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

The assembler message at the bottom right states: "Program assembled successfully".

13) Odd/Even

The screenshot shows the GNUSim8085 - 8085 Microprocessor Simulator interface. The assembly code is as follows:

```
1 LDA 2500
2 ANI 01H
3 JZ EVEN
4 MVI A,01H
5 STA 2501
6 HLT
7 EVEN: MVI A,00H
8 STA 2501
9 HLT
```

The registers section shows the following values:

Register	Value
A	01
BC	32 28
DE	00 00
HL	00 00
PSW	00 00
PC	42 0E
SP	FF FF
Int-Reg	00

The flags section shows the following values:

Flag	Value
S	0
Z	0
AC	1
P	0
C	0

The memory section shows the following values:

Address (Hex)	Address	Data
09C4	2500	35
09C5	2501	1
09C6	2502	5
09C7	2503	0
09C8	2504	0
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

The assembler message shows: "Program assembled successfully".

14) Positive/Negative

The screenshot shows the GNUSim8085 - 8085 Microprocessor Simulator interface. The assembly code is as follows:

```
1 LDA 2500
2 ORA A
3 JP POSITIVE
4 MVI A,01H
5 STA 2501
6 HLT
7 POSITIVE: MVI A,00H
8 STA 2501
9 HLT
```

The registers section shows the following values:

Register	Value
A	00
BC	32 28
DE	00 00
HL	00 00
PSW	00 00
PC	42 13
SP	FF FF
Int-Reg	00

The flags section shows the following values:

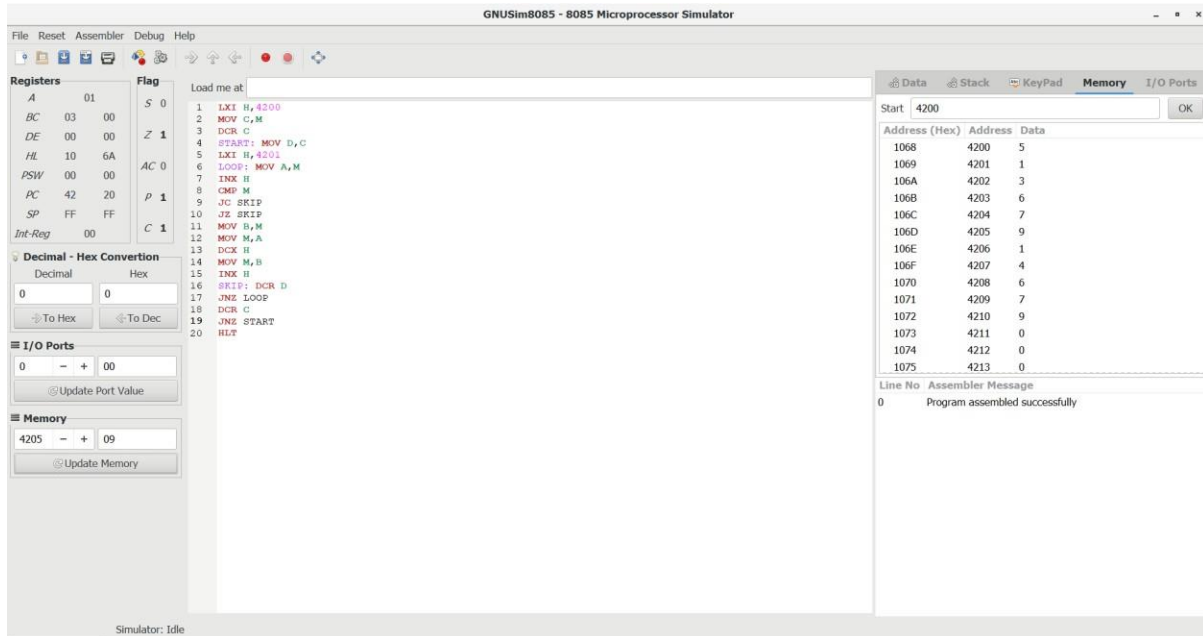
Flag	Value
S	0
Z	0
AC	0
P	0
C	0

The memory section shows the following values:

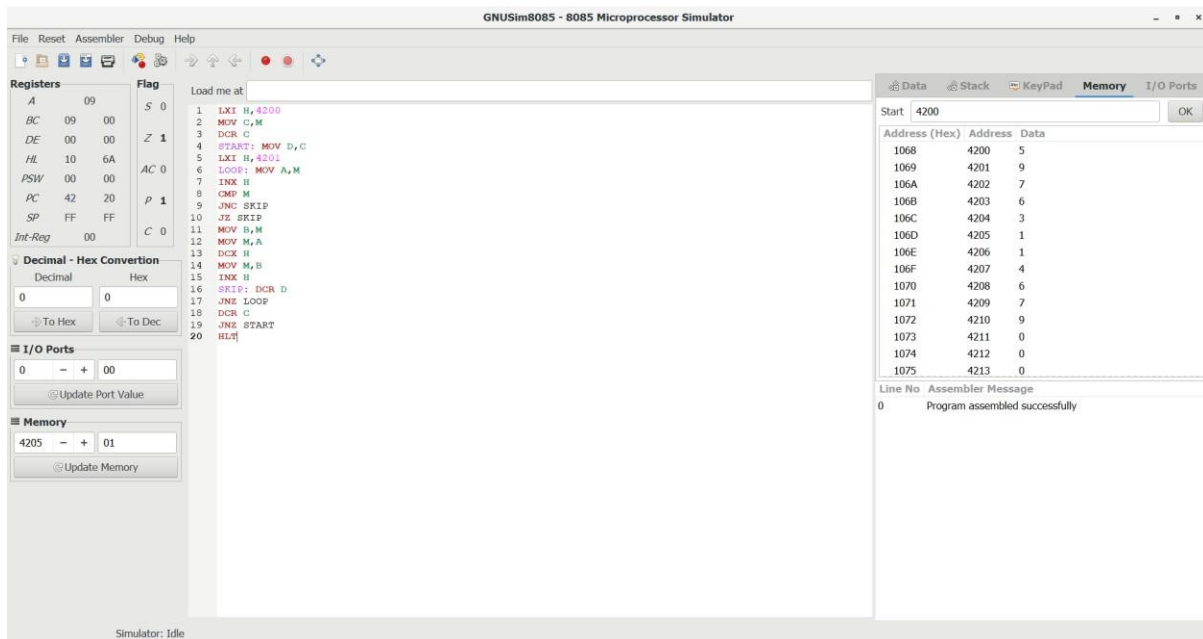
Address (Hex)	Address	Data
09C4	2500	25
09C5	2501	0
09C6	2502	5
09C7	2503	0
09C8	2504	0
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

The assembler message shows: "Program assembled successfully".

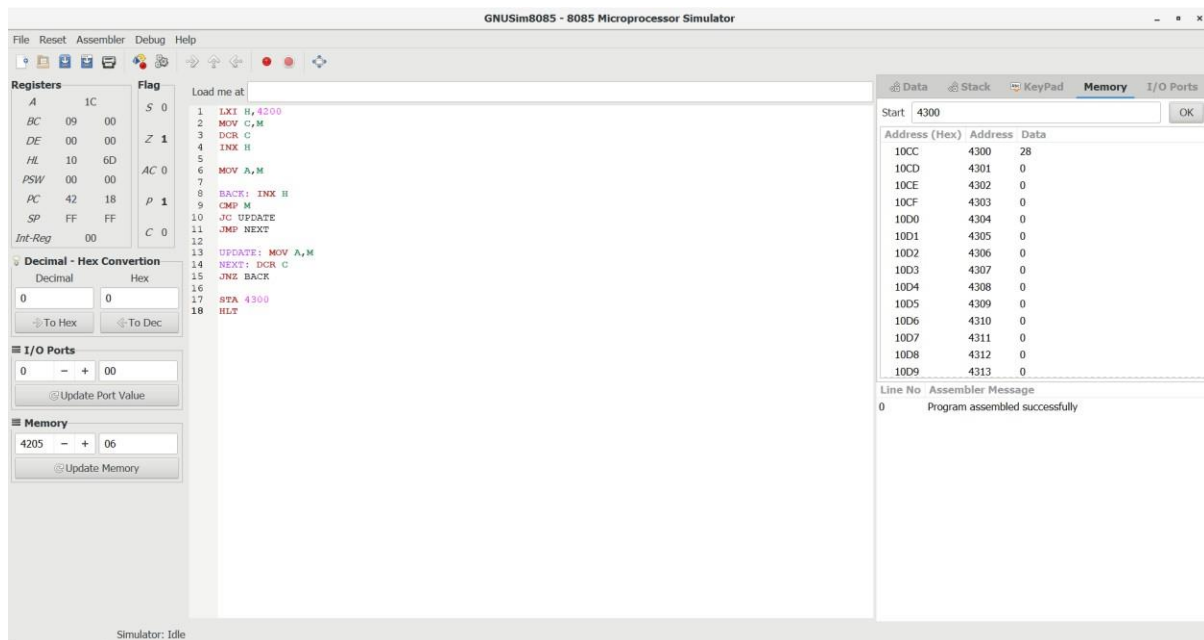
15) Ascending order



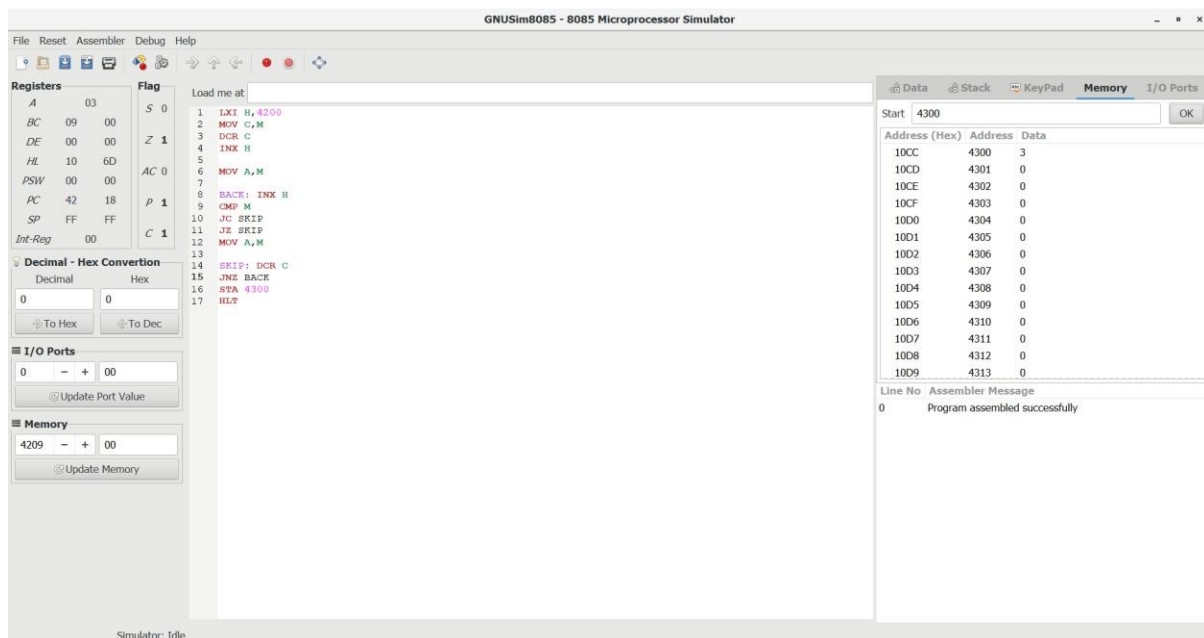
16) Descending order.



17) largest number in an array.



18) Minimum number in an array.



19) LCM of Two 8-bit Numbers

The screenshot shows the GNUSim8085 - 8085 Microprocessor Simulator interface. The main window displays the assembly code for calculating the LCM of two 8-bit numbers. The code is as follows:

```
1 LDA 4200
2 STA 4300
3
4 LDA 4201
5 MOV B,A
6
7 CHECK: LDA 4300
8 MOV D,A
9 DIVIDE: MOV A,D
10 CMP E
11 JC NOT_DIVISIBLE
12 JZ FOUND
13 SUB E
14 MOV D,A
15 JMP DIVIDE
16 NOT_DIVISIBLE: LDA 4300
17 MOV B,A
18 LDA 4200
19 ADD B
20 STA 4300
21 JMP CHECK
22 FOUND: HLT
```

The registers section shows the following values:

Register	Value
A	02
BC	03 00
DE	02 02
HL	00 00
PSW	00 00
PC	42 2A
SP	FF FF
Int-Reg	00

The flags section shows the following values:

Flag	Value
S	0
Z	1
AC	0
P	1
C	0

The memory section shows the following values:

Address (Hex)	Address	Data
10CC	4300	6
10CD	4301	0
10CE	4302	0
10CF	4303	0
10D0	4304	0
10D1	4305	0
10D2	4306	0
10D3	4307	0
10D4	4308	0
10D5	4309	0
10D6	4310	0
10D7	4311	0
10D8	4312	0
10D9	4313	0

The simulator status bar at the bottom indicates "Simulator: Idle".

20) GCD of Two 8-bit Numbers

The screenshot shows the GNUSim8085 - 8085 Microprocessor Simulator interface. The main window displays the assembly code for calculating the GCD of two 8-bit numbers. The code is as follows:

```
1 LDA 4200
2 MOV B,A
3 LDA 4201
4 MOV C,A
5 LOOP: MOV A,B
6 ORA A
7 JZ BZERO
8 MOV A,C
9 ORA A
10 JZ CZERO
11 MOV A,B
12 CMP C
13 JC CGREATER
14 JZ EQUAL
15 SUB C
16 MOV B,A
17 JMP LOOP
18 CGREATER: MOV A,C
19 SUB B
20 MOV C,A
21 JMP LOOP
22 EQUAL: MOV A,B
23 STA 4205
24 HLT
25 BZERO: MOV A,C
26 STA 4205
27 HLT
28 CZERO: MOV A,B
29 STA 4205
30 HLT
```

The registers section shows the following values:

Register	Value
A	01
BC	01 01
DE	02 02
HL	00 00
PSW	00 00
PC	42 2A
SP	FF FF
Int-Reg	00

The flags section shows the following values:

Flag	Value
S	0
Z	1
AC	0
P	1
C	0

The memory section shows the following values:

Address (Hex)	Address	Data
1068	4200	3
1069	4201	2
106A	4202	0
106B	4203	0
106C	4204	0
106D	4205	1
106E	4206	0
106F	4207	0
1070	4208	0
1071	4209	0
1072	4210	0
1073	4211	0
1074	4212	0
1075	4213	0

The simulator status bar at the bottom indicates "Simulator: Idle".

21) Factorial of an 8-bit Number

GNUSim8085 - 8085 Microprocessor Simulator

File Reset Assembler Debug Help

Registers: A 78, BC 78 00, DE 06 78, HL 00 00, PSW 00 00, PC 42 28, SP FF FF, Int-Reg 00. Flag: S 1, Z 0, AC 0, P 1, C 1.

Decimal - Hex Conversion: Decimal 0, Hex 0. To Hex, To Dec.

I/O Ports: 0, 00. Update Port Value.

Memory: 4200, 05. Update Memory.

Load me at: 1 LDA 4200, 2 MOV C,A, 3, 4 MVI B,01, 5 MVI D,02, 6, 7 CHECK: LDA 4200, 8 CMP D, 9 JC DONE, 10 JZ MULTIPLY, 11, 12 MULTIPLY: MOV A,D, 13 MOV C,A, 14 MVI B,00, 15, 16 ADD_LOOP: MOV A,E, 17 ADD B, 18 MOV E,A, 19 DCR C, 20 JNZ ADD_LOOP, 21, 22 MOV A,E, 23 MOV B,A, 24, 25 INR D, 26 JMP CHECK, 27, 28 DONE: MOV A,B, 29 STA 4300, 30 HLT.

Memory Window: Start 4300. Address (Hex) Address Data. 10CC 4300 120, 10CD 4301 0, 10CE 4302 0, 10CF 4303 0, 10D0 4304 0, 10D1 4305 0, 10D2 4306 0, 10D3 4307 0, 10D4 4308 0, 10D5 4309 0, 10D6 4310 0, 10D7 4311 0, 10D8 4312 0, 10D9 4313 0.

Line No Assembler Message: 0 Program assembled successfully.

Simulator: Idle

22) Convert decimal into hexadecimal

GNUSim8085 - 8085 Microprocessor Simulator

File Reset Assembler Debug Help

Registers: A A9, BC 90 09, DE A0 00, HL 00 00, PSW 00 00, PC 42 20, SP FF FF, Int-Reg 00. Flag: S 1, Z 0, AC 0, P 1, C 0.

Decimal - Hex Conversion: Decimal 0, Hex 0. To Hex, To Dec.

I/O Ports: 0, 00. Update Port Value.

Memory: 4200, 19. Update Memory.

Load me at: 1 LDA 4200, 2 MOV B,A, 3, 4 ANI 0FH, 5 MOV C,A, 6, 7 MOV A,B, 8 ANI 0FH, 9, 10 RRC, 11 RRC, 12 RRC, 13 RRC, 14, 15 MOV B,A, 16, 17 MVI D,00, 18 MVI E,0AH, 19, 20 TEN_TIMES: MOV A,D, 21 ADD B, 22 MOV D,A, 23 DCR E, 24 JNZ TEN_TIMES, 25 MOV A,D, 26 ADD C, 27 STA 4205, 28, 29 HLT.

Memory Window: Start 4200. Address (Hex) Address Data. 1068 4200 25, 1069 4201 2, 106A 4202 0, 106B 4203 0, 106C 4204 0, 106D 4205 169, 106E 4206 0, 106F 4207 0, 1070 4208 0, 1071 4209 0, 1072 4210 0, 1073 4211 0, 1074 4212 0, 1075 4213 0.

Line No Assembler Message: 0 Program assembled successfully.

Simulator: Idle